

Workshop 2.2: Electrostatic Analysis

Release 2020R2



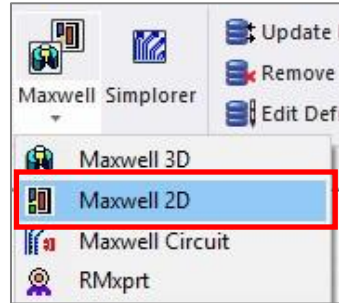
Overview

- Introduction to the Electrostatic Solver
 - This workshop introduces the Electro Static solver based on some simple examples. This solver is meant to solve the static electric field without current flowing in conductors (conductors are in electrostatic equilibrium). The conductors are considered perfect such that there is no electric field inside them
- The Workshop contains following three examples
 - Example1: **Cylindrical Capacitor in RZ**
 - In this example, we will determine the electric field distribution of coaxial cable based on the potential (or the charges) that are applied on each conductor. Coaxial cable will be solved with RZ representation
 - Example2: **Cylindrical Capacitor in XY**
 - The same problem in XY representation
 - Example3: **Capacitance of a Planar Capacitor**
 - In this example we simulate a simple planar capacitor made of two parallel conducting plates

Example 1: Cylindrical Capacitor RZ - Model Setup

- Insert Design

- Select the menu item **Project** → **Insert Maxwell 2D Design**, or click on the icon in drop down list Maxwell on panel Desktop

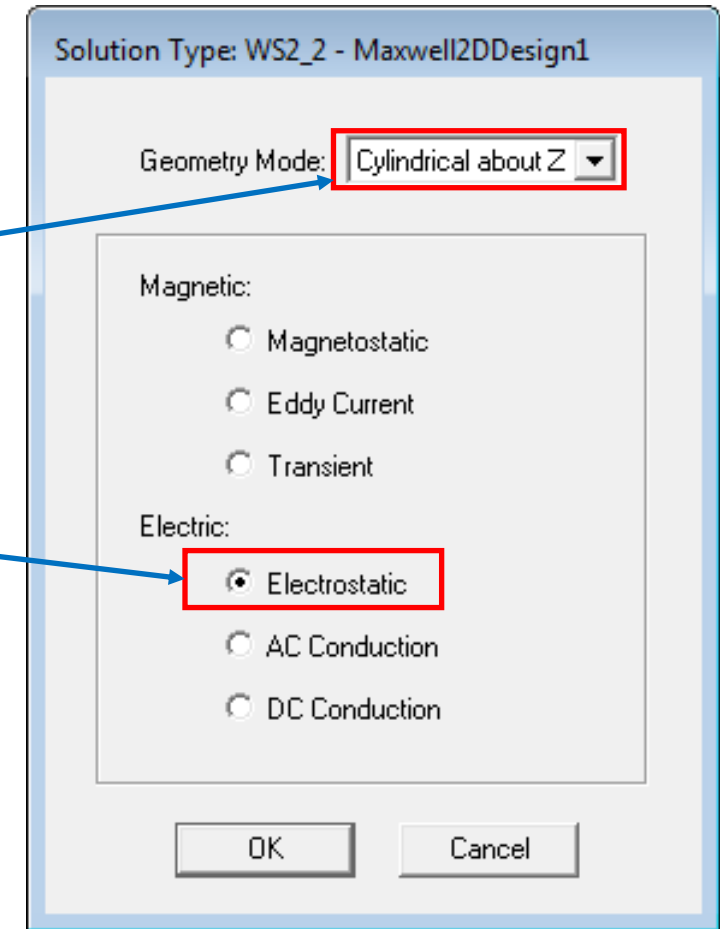


- Set Solution Type

- Select the menu item **Maxwell 2D** → **Solution Type**
- Geometry Mode: **Cylindrical about Z**
- Choose **Electric** → **Electrostatic**
- Click the OK button

- Set Model Units

- Select the menu item **Modeler** → **Units**
- In Set Modeler Units window,
 - Select units: mm (millimeters)
 - Press the OK button



Create Model

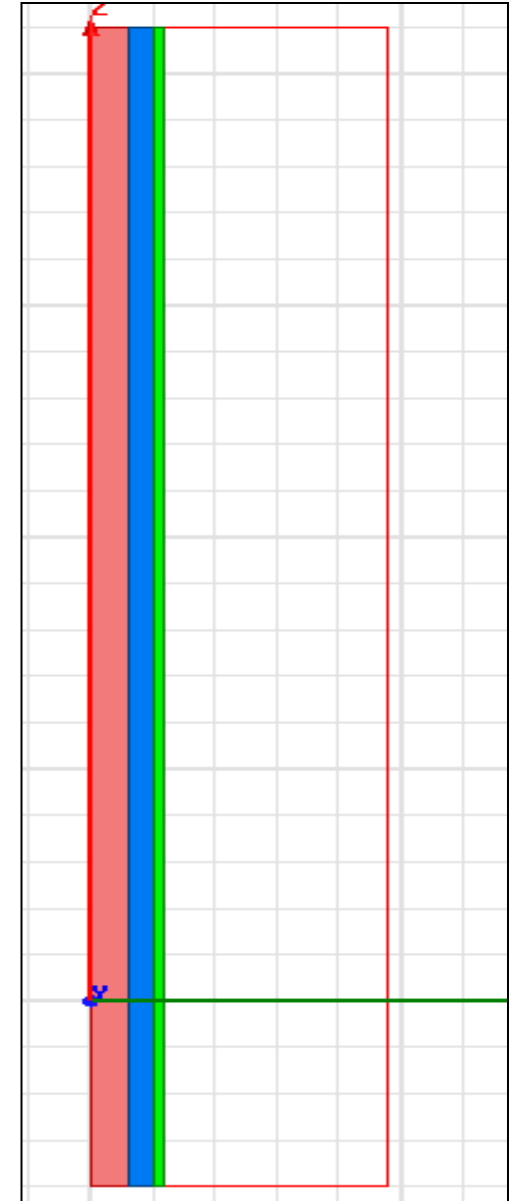
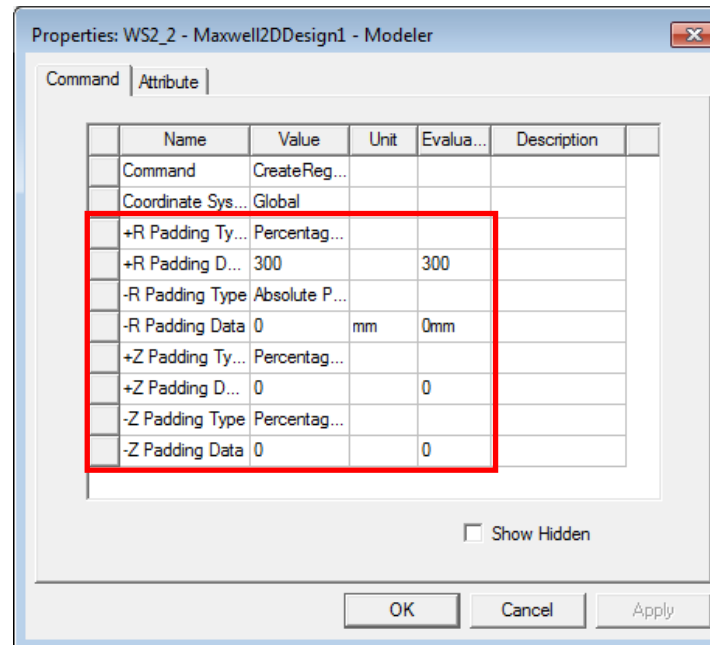
- Create object **Inner**
 - Select the menu item **Draw → Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
X: 0, Y: 0, Z: -4, Press the Enter key
 - Using the coordinate entry fields, enter the opposite corner
dX: 0.6, dY: 0, dZ: 25, Press the Enter key
 - Change the name of resulting sheet to **Inner** and color to **Light Red**
 - Change the material of the object to **Copper**
- Create Object **Airgap**
 - Select the menu item **Draw → Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
X: 0.6, Y: 0, Z: -4, Press the Enter key
 - Using the coordinate entry fields, enter the opposite corner
dX: 0.4, dY: 0, dZ: 25, Press the Enter key
 - Change the name of resulting sheet to **Airgap** and color to **Light Blue**
 - Change the material of the object to **Air**

Name	Value	Unit	Evaluated Value
Name	Inner		
Material	"copper"		"copper"
Solve Inside	☒		
Orientation	Global		
Model	☒		
Display Wi...	☐		
Color			
Transparent	0		

Name	Value	Unit	Evaluated Value
Name	Airgap		
Material	"air"		"air"
Solve Inside	☒		
Orientation	Global		
Model	☒		
Display Wi...	☐		
Color			
Transparent	0		

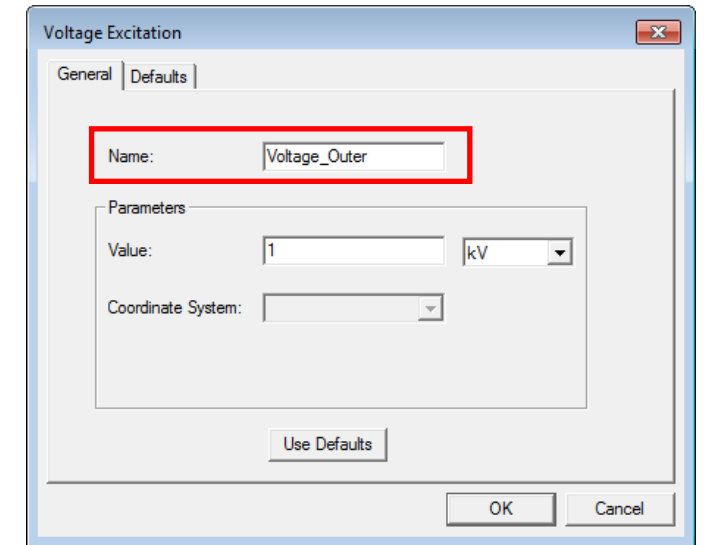
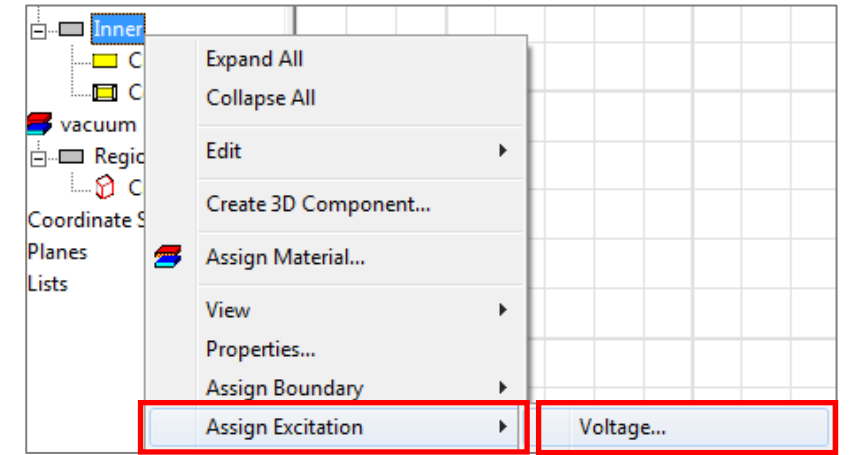
Create Model

- Create object **Outer**
 - Select the menu item **Draw** → **Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
X: 1, Y: 0, Z: -4, Press the **Enter** key
 - Using the coordinate entry fields, enter the opposite corner
dX: 0.2, dY: 0, dZ: 25, Press the **Enter** key
 - Change the name of resulting sheet to **Outer** and color to **Light green**
 - Change the material of the object to **Copper**
- Create Solution Region
 - Select the menu item **Draw** → **Region**
 - Pad individual directions: ☒ Checked
 - Padding Type: Percentage Offset
 - +R = 300
 - Specify rest to 0
 - Press OK



Assign Excitations

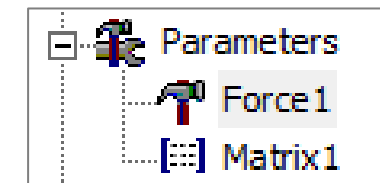
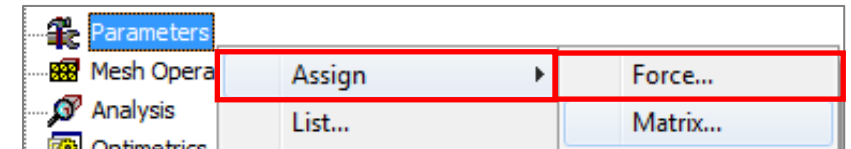
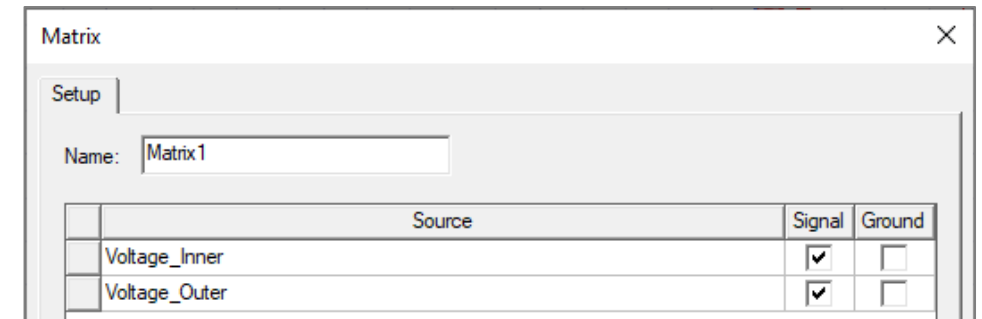
- Assign Excitation to object Inner
 - Select the sheet **Inner** from the history tree
 - Select the menu **Maxwell 2D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Name: Voltage_Inner
 - Value: -1kV
 - Press OK
- Assign Excitation to object Outer
 - Select the sheet **Outer** from the history tree
 - Select the menu **Maxwell 2D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Name: Voltage_Outer
 - Value: 1kV
 - Press OK



Note: Assuming an electrostatic equilibrium, the conductors are assigned as an equipotential voltage

/ Assign Parameters

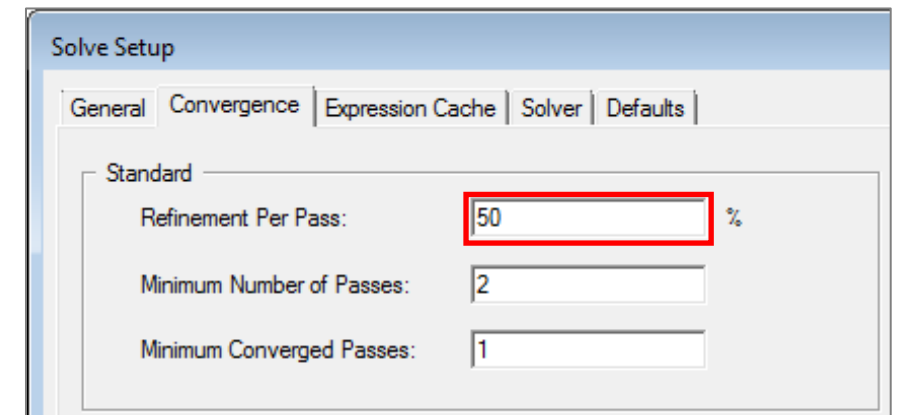
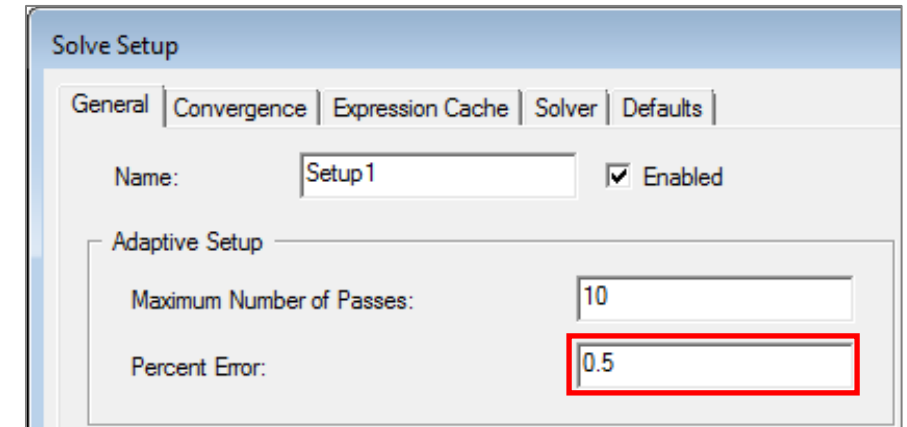
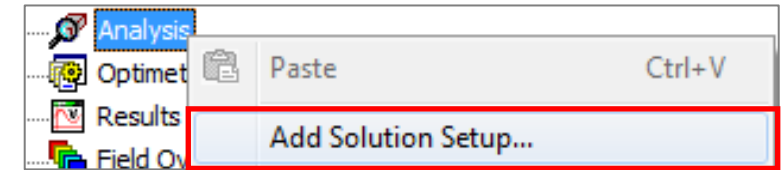
- Assign Capacitance Computation
 - From the Project Manager *RMB on Parameters* → *Assign* → *Matrix*
 - In Matrix window, for all the sources
 - Signal Line: ☒ Checked
 - Press **OK**
- Assign Force Computation
 - Select the sheet **Inner** from history tree
 - From the Project *Manager RMB on Parameters* → *Assign* → *Force*
 - In Force Setup window, press OK



Note: In addition to the fields, we are interested on the Capacitance values as well as the force applied to the inner armature.

Analyze

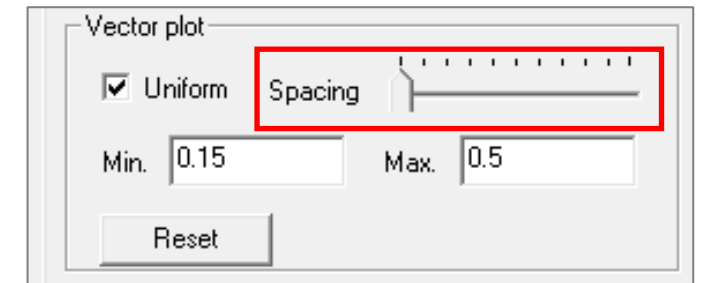
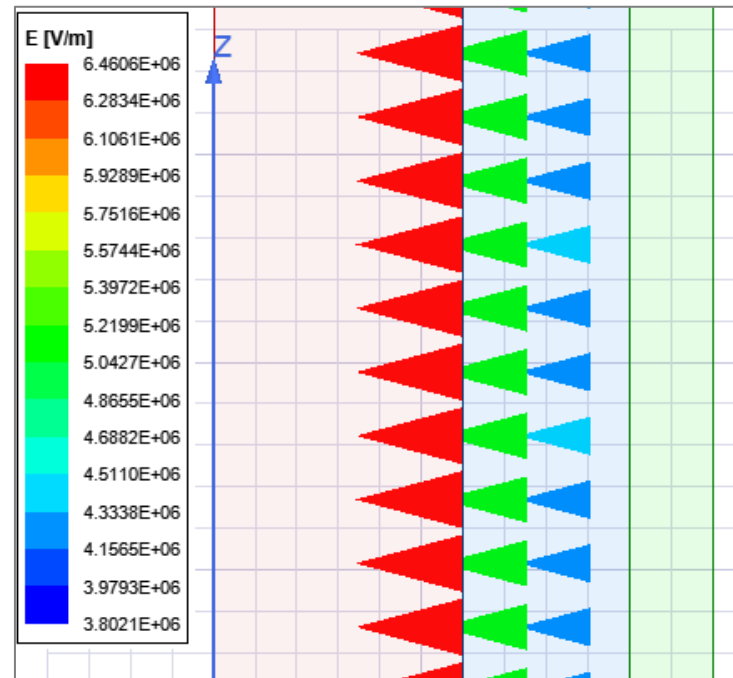
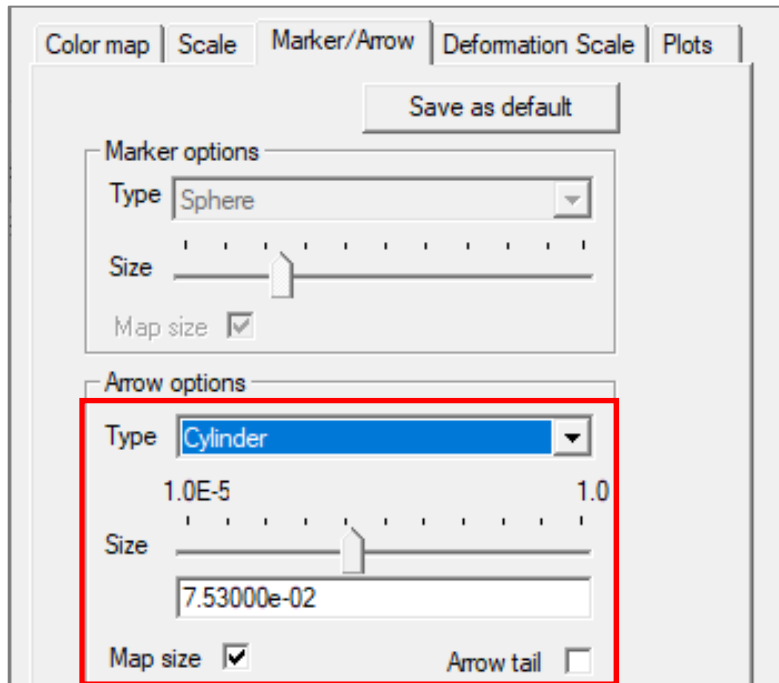
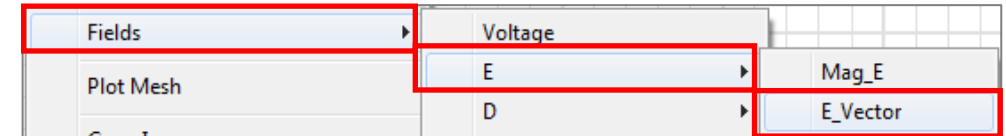
- Create an analysis setup:
 - Select the menu item **Maxwell 2D** → **Analysis Setup** → **Add Solution Setup**
 - General Tab
 - Percentage Error: 0.5
 - Convergence Tab
 - Refinement Per Pass: 50%
 - Click the OK button
- Start the solution process:
 - In the Project Manager window **RMB on Setup1** → **Analyze**



Plot Electric Field Vectors

- Plot Electric Field Vectors

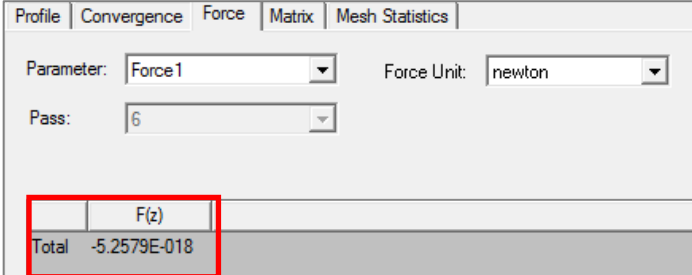
- Select the object **Air** from history tree
- Select the menu **Maxwell 2D** → **Fields** → **Fields** → **E** → **E_Vector**
- In Create Field Plot window, press **Done**
- To adjust arrows size and spacing, double click on the legend and go to Marker/Arrow and Plots tabs



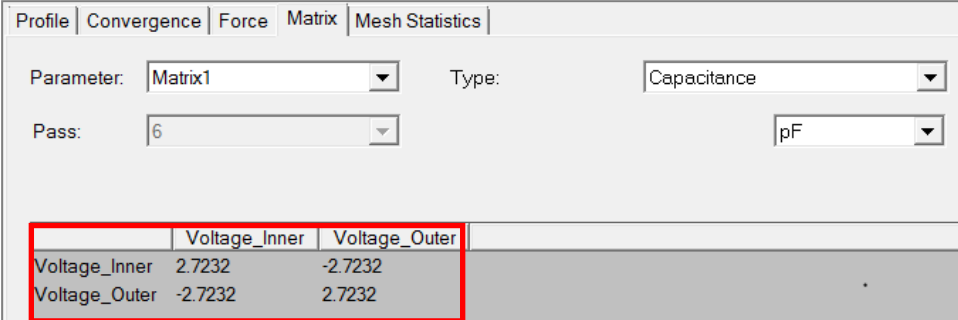
View Results

- View Results

- Select the menu *Maxwell 2D → Results → Solution Data*
- Select Force tab
 - Note: The force is practically zero since the model is electrically balanced.
- Select Matrix tab



	F(z)
Total	-5.2579E-018



	Voltage_Inner	Voltage_Outer
Voltage_Inner	2.7232	-2.7232
Voltage_Outer	-2.7232	2.7232

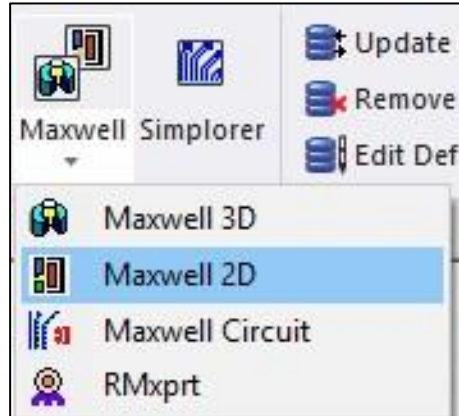
- The analytical value of the capacitance per meter is given by the following formula:
 $C = 2\pi\epsilon_0 / \ln(b/a)$ (a and b being the inside and outside diameters)
- The analytical value is therefore 1.089e-10 Farad/meter (a = 0.6 mm, b = 1 mm)
- In our project, length of the conductor is 25 mm, therefore the total capacitance is 2.723pF. We have a good agreement with the obtained result 2.7245 pF.

Note: in the Convergence tab, we find the total energy of the system, 5.4489e-6 J. It is exactly 20002/2 times the capacitance (2000V being the difference of potential).

Example 2: Cylindrical Capacitor XY – Model setup

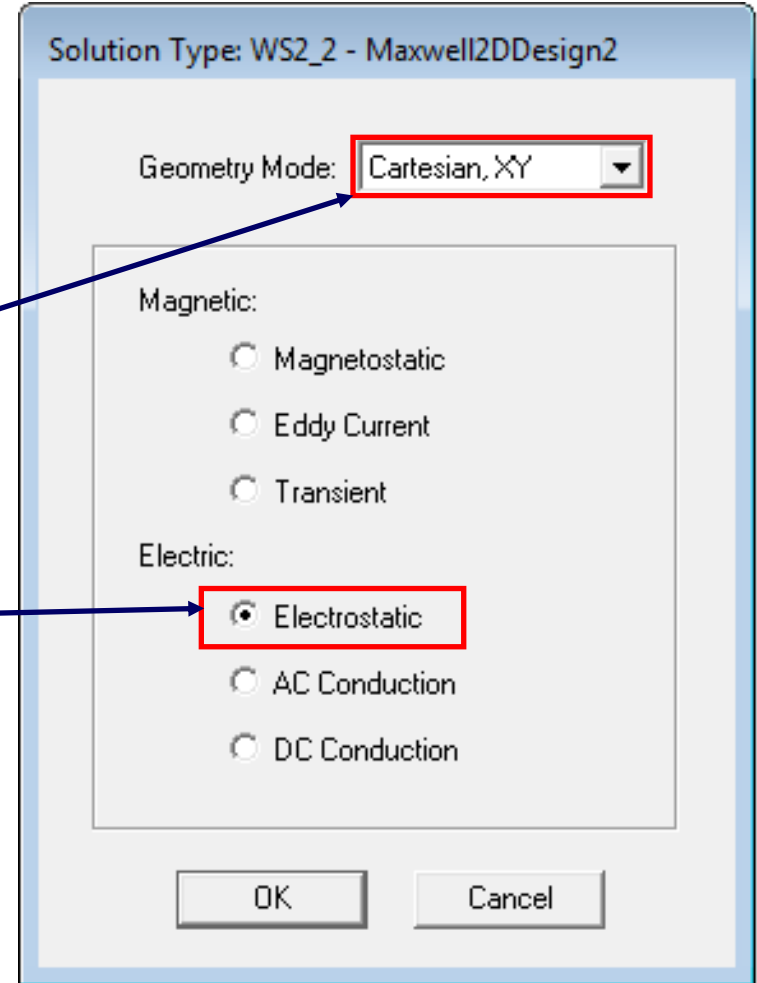
- Insert Design

- Select the menu item **Project** → **Insert Maxwell 2D Design**, or click on the icon in drop down list Maxwell on panel Desktop



- Set Solution Type

- Select the menu item **Maxwell 2D** → **Solution Type**
- Geometry Mode: **Cartesian XY**
- Choose **Electric** → **Electrostatic**
- Click the OK button

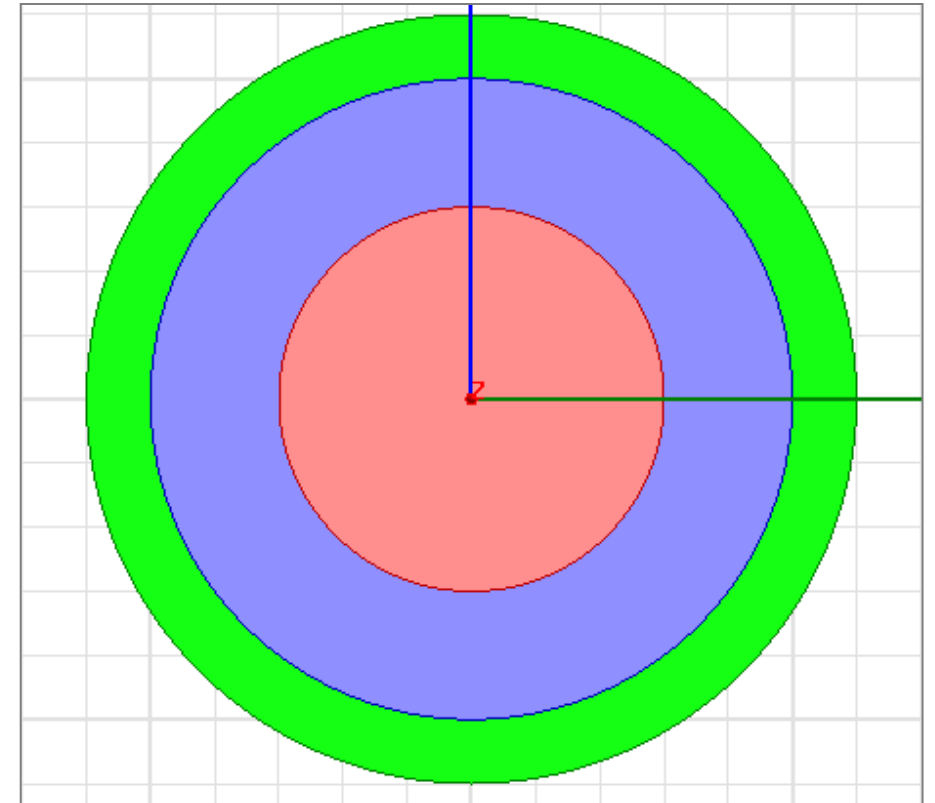


Create Model

- Create object **Inner**
 - Select the menu item **Draw → Circle**
 - Using the coordinate entry fields, enter the center of the circle
X: 0, Y: 0, Z: 0, Press the Enter key
 - Using the coordinate entry fields, enter the radius
dX: 0.6, dY: 0, dZ: 0, Press the Enter key
 - Change the name of resulting sheet to **Inner** and color to **Light Red**
 - Change the material of the object to **Copper**
- Create **Airgap**
 - Select the menu item **Draw → Circle**
 - Using the coordinate entry fields, enter the center of the circle
X: 0, Y: 0, Z: 0, Press the Enter key
 - Using the coordinate entry fields, enter the radius
dX: 1, dY: 0, dZ: 0, Press the Enter key
 - Change the name of resulting sheet to **Airgap** and color to **Light Blue**
 - Change the material of the object to **Air**

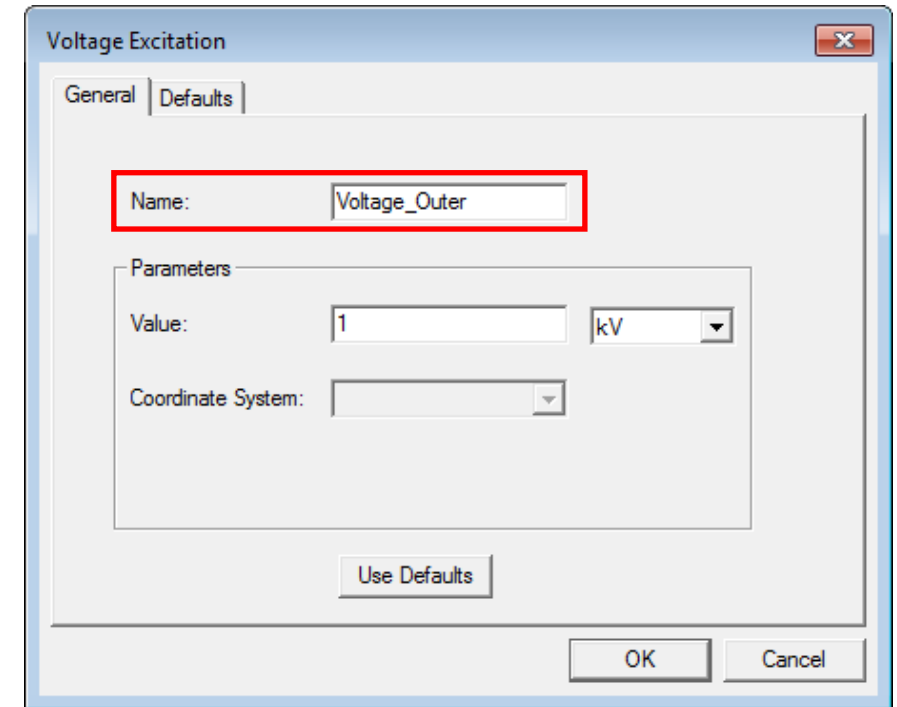
Create Model

- Create Object **Outer**
 - Select the menu item **Draw → Circle**
 - Using the coordinate entry fields, enter the center of the circle
X: 0, Y: 0, Z: 0, Press the Enter key
 - Using the coordinate entry fields, enter the radius
dX: 1.2, dY: 0, dZ: 0, Press the Enter key
 - Change the name of resulting sheet to **Outer** and color to **Light Green**
 - Change the material of the object to **Copper**



/ Assign Excitations

- Assign Excitation to object **Inner**
 - Select the sheet **Inner** from the history tree
 - Select the menu item **Maxwell 2D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Name: Voltage_Inner
 - Value: -1kV
 - Press OK
- Assign Excitation to object **Outer**
 - Select the sheet **Outer** from the history tree
 - Select the menu item **Maxwell 2D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Name: Voltage_Outer
 - Value: 1kV
 - Press OK



Note: Assuming an electrostatic equilibrium, the conductors are assigned as an equipotential voltage

/ Assign Parameters

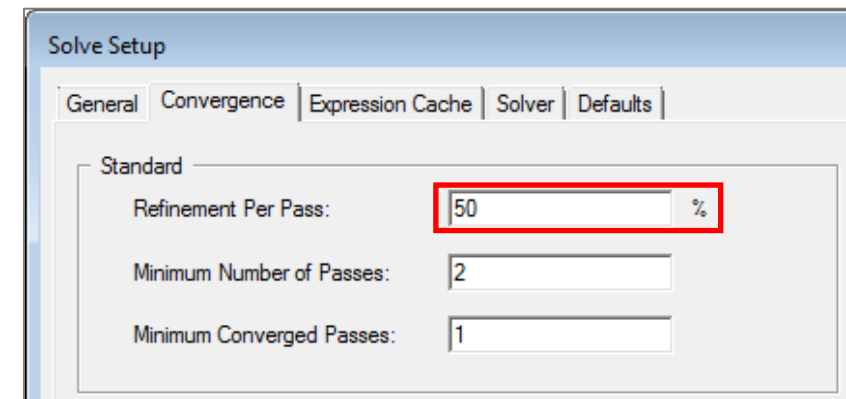
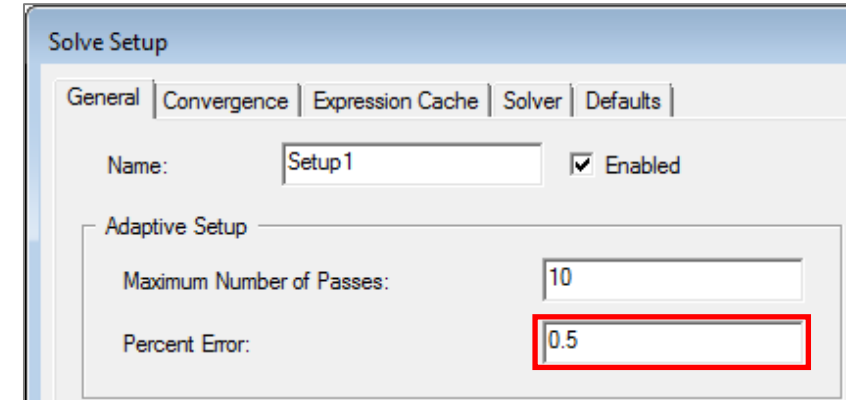
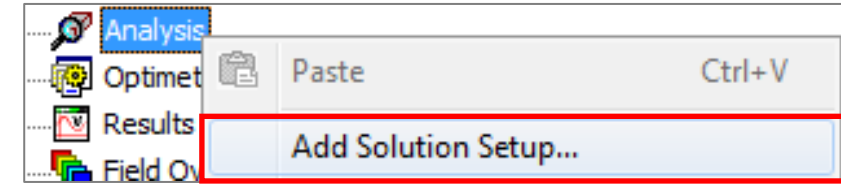
- Assign Capacitance Computation
 - From the Project Manager window *RMB on Parameters* → *Assign* → *Matrix*
 - In Matrix window, for **Voltage_inner**
 - Signal Line: ☒ Checked
 - In Matrix window, for **Voltage_outer**
 - Ground: ☒ Checked
 - Press OK
 - We ground Voltage_Outer to obtain just a 1 by 1 matrix.
- Assign Force Computation
 - Select the sheet **Inner** from history tree
 - Select the menu item *Maxwell 2D* → *Parameters* → *Assign* → *Force*
 - In Force Setup window, press OK

	Source	Signal	Ground
	Voltage_Inner	☒	☐
	Voltage_Outer	☐	☒

Note: In addition to the fields, we are interested on the Capacitance values as well as the force applied to the inner armature

Analyze

- Create an analysis setup:
 - Select the menu item **Maxwell 2D** → **Analysis Setup** → **Add Solution Setup**
 - General Tab
 - Percentage Error: 0.5
 - Convergence Tab
 - Refinement Per Pass: 50%
 - Click the OK button
- Start the solution process:
 - In the Project Manager **window RMB on Setup1** → **Analyze**



View Results

- View Results

- Select the menu *Maxwell 2D → Results → Solution Data*
- Select Force tab
 - **Note:** the force is practically zero since the model is electrically balanced
- Select Matrix tab
- The analytical formula $C = 2\pi\epsilon_0 / \ln(b/a)$ gives a value of $1.089\text{e-}10$ Farad/meter
- This matches the obtained value

Parameter:	Force1	Force Unit:	newton
Pass:	2		
	F(x)	F(y)	Mag(F)
Total	-8.5498E-005	-2.5406E-005	8.9193E-005

Simulation: Setup1 LastAdaptive

Design Variation:

Profile Convergence Force Torque Matrix Mesh Statistics

Parameter: Matrix1 Type: Capacitance

Pass: 2 Capacitance Units: pF

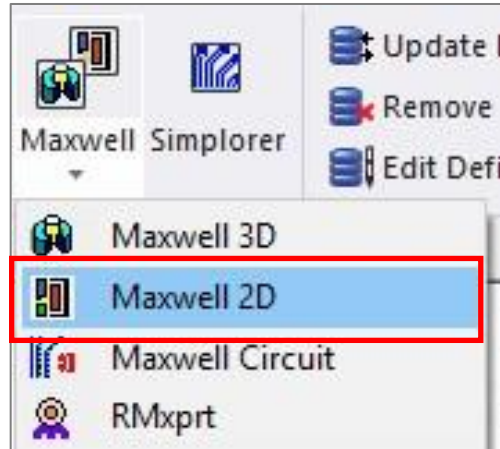
	Voltage_Inner
Voltage_Inner	108.98

Note: In 2DXY Geometry all the above reported results are per-meter length, i.e. the model is considered to have 1 m depth in Z-direction

Example 3: Planar Capacitor XY – Model setup

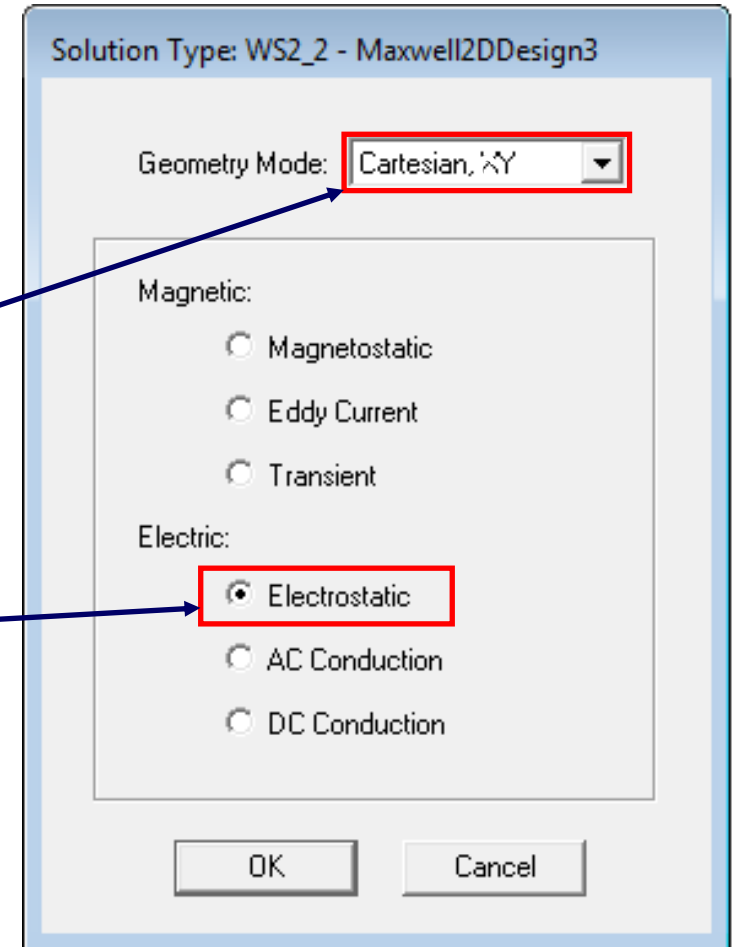
- Insert Design

- Select the menu item **Project** → **Insert Maxwell 2D Design**, or click on the icon in drop down list Maxwell on panel Desktop



- Set Solution Type

- Select the menu item **Maxwell 2D** → **Solution Type**
- Geometry Mode: **Cartesian XY**
- Choose **Electric** → **Electrostatic**
- Click the OK button

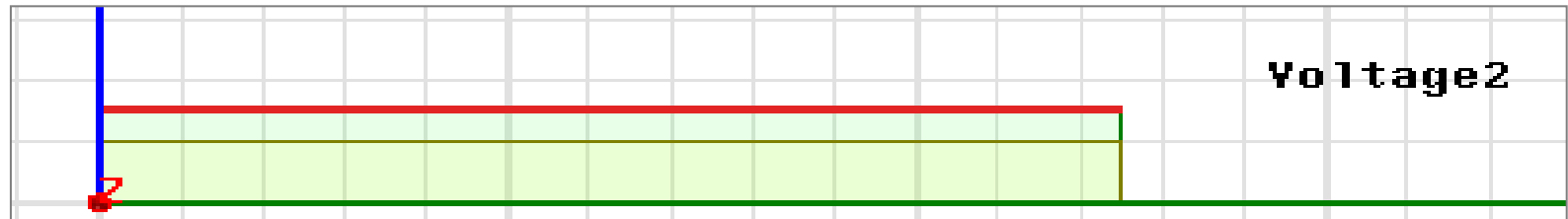


Create Model

- Create object **DownPlate**
 - Select the menu item **Draw → Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
X: 0, Y: 0, Z: 0, Press the Enter key
 - Using the coordinate entry fields, enter the opposite corner
dX: 25, dY: 2, dZ: 0, Press the Enter key
 - Change the name of resulting sheet to **DownPlate** and color to **Yellow**
 - Change the material of the object to **Copper**
- Create **Region**
 - Select the menu item **Draw → Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
X: 0, Y: 0, Z: 0, Press the Enter key
 - Using the coordinate entry fields, enter the opposite corner
dX: 25, dY: 3, dZ: 0, Press the Enter key
 - Change the name of resulting sheet to **Region** and color to **Green**
 - Change the material of the object to **Air**

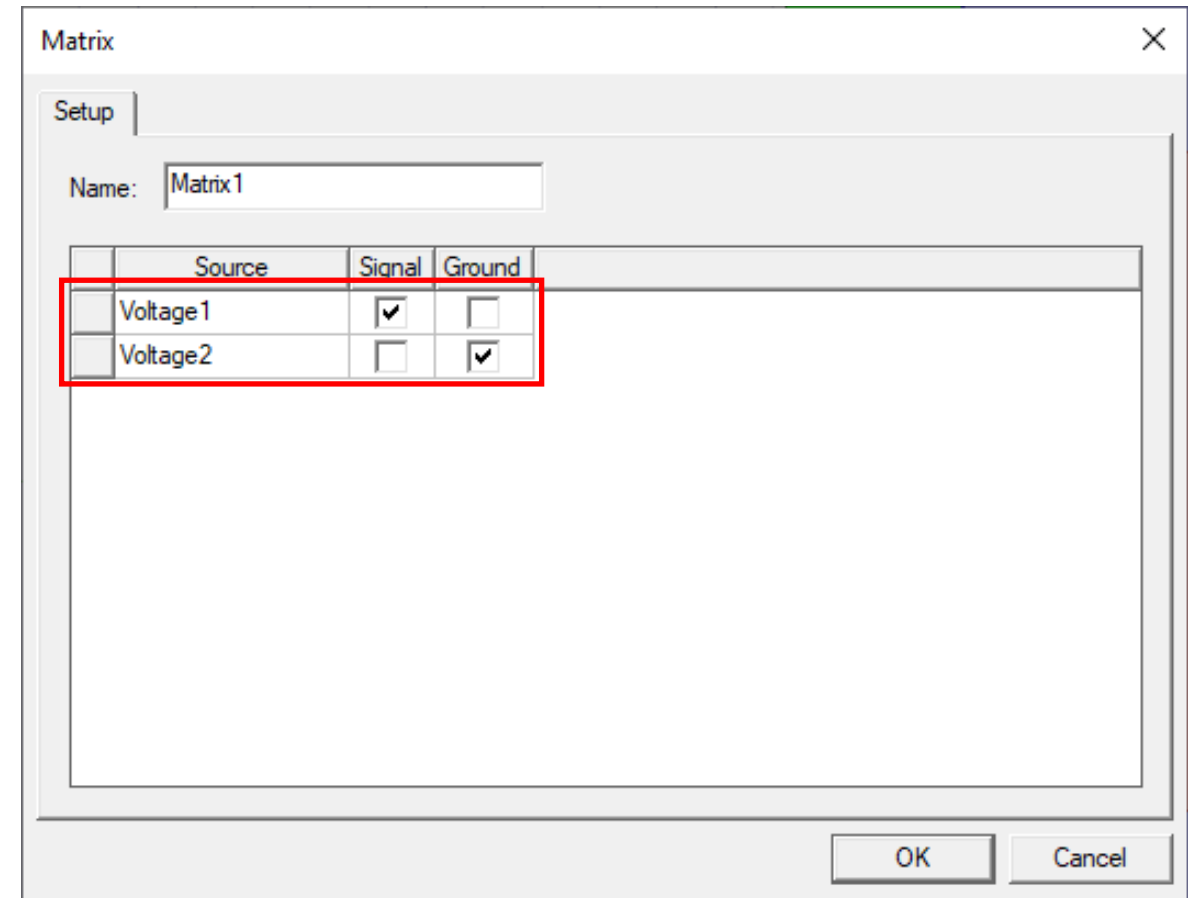
/ Assign Excitations

- Assign Excitation to object DownPlate
 - Select the sheet **Downplate** from the history tree
 - Select the menu item **Maxwell 2D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Value: 0 V
 - Press OK
- Assign Excitation to object Outer
 - Select the menu item Edit → Select → Edges
 - Select the top edge of the Region as shown in below image
 - Select the menu item **Maxwell 2D → Excitations → Assign → Voltage**
 - In Voltage Excitation window,
 - Value: 1 V
 - Press OK



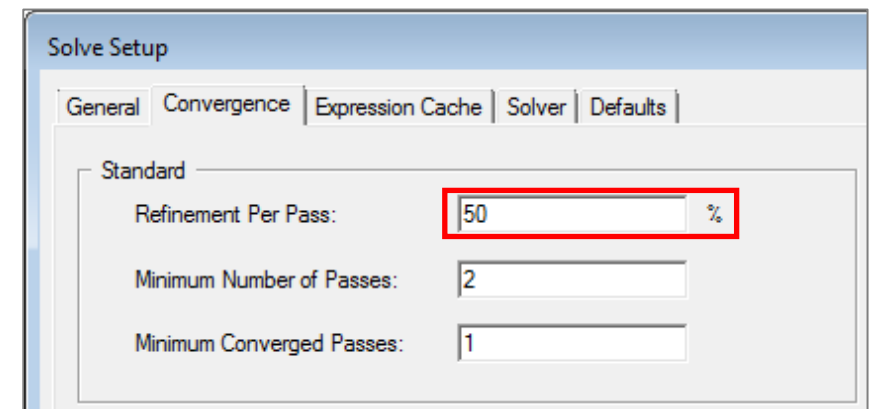
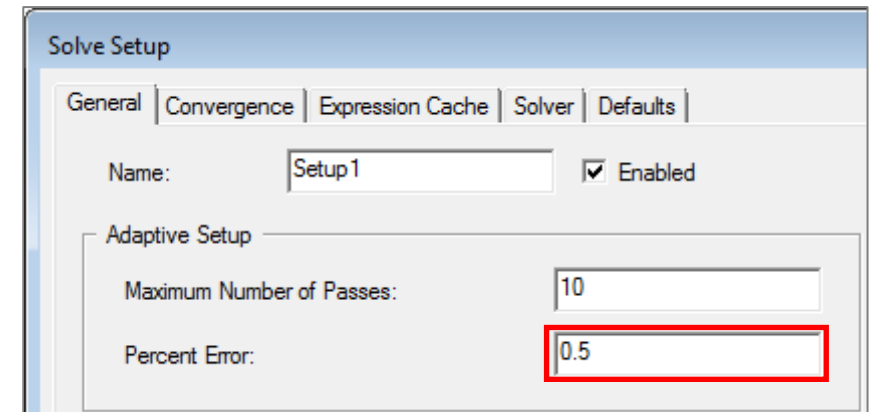
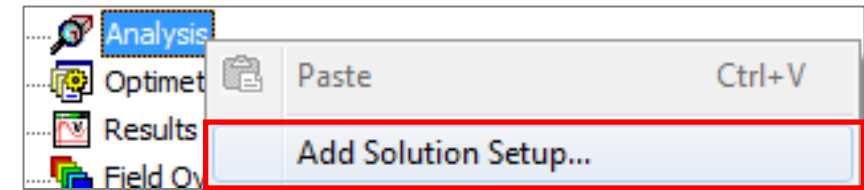
/ Assign Parameters

- Assign Capacitance Computation
 - From the Project Manager *RMB on Parameters* → *Assign* → *Matrix*
 - In Matrix window for Voltage 1
 - Signal Line: ☒ Checked
 - In Matrix window for Voltage 2
 - Ground: ☒ Checked
 - Press OK



Analyze

- Create an analysis setup:
 - Select the menu item **Maxwell 2D** → **Analysis Setup** → **Add Solution Setup**
 - General Tab
 - Percentage Error: 0.5
 - Convergence Tab
 - Refinement Per Pass: 50%
 - Click the OK button
- Start the solution process:
 - In the Project Manager window **RMB on Setup1** → **Analyze**



View Results

- View Capacitance
 - Select the menu *Maxwell 2D* → *Results* → *Solution Data*
 - In Solutions window
 - Select Matrix tab

Simulation: Setup1 LastAdaptive

Design Variation:

Profile | Convergence | Force | Torque | **Matrix** | Mesh Statistics

Parameter: Matrix1 Type: Capacitance

Pass: 2 Capacitance Units: pF

	Voltage1
Voltage1	221.49

- The analytical value of the capacitance for two parallel plates is given by
$$C = A/d * \epsilon_0$$
(**A** is the area of the plate and **d** is the thickness of the dielectric)
- If we consider the plate to be 25mm by 25 mm, using the above formula, we obtain **5.53 pF**
- The obtained value (221.49pF) is valid for the two parallel plates with a 1 meter depth. Rescaling this value by multiplying by 0.025m it gives **5.53pF** as well

Saving the Project

- This completes the workshop
- Save the file with the name **Workshop_2_2** in the working folder



End of Presentation